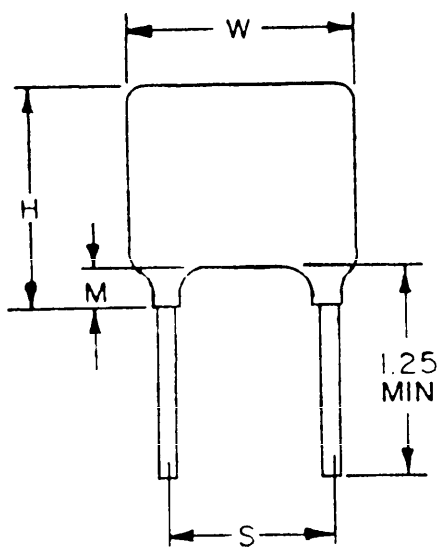
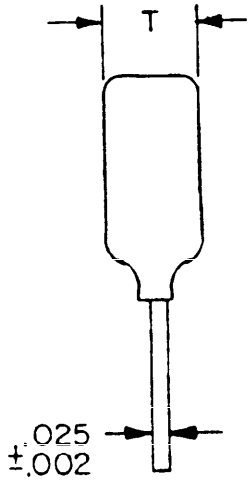
MILITARY SPECIFICATION SHEET

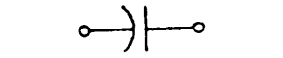
Capacitors, Ceramic, Multilayer, High Voltage, 4000 V DC, Established Reliability, Style HVR04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49467.







CIRCUIT DIAGRAM

Case code	Dimensions				
	W max	H max	T max	M max	S ±.030
D	.470	.400	.270	.060	.375
E	.570	.500	.270	.060	.475
F	.670	.600	.270	.060	.575
G	.770	.720	.270	.060	.675
J	1.300	.600	.270	.060	1.175
K	1.500	.720	.270	.060	1.375

Inches	mm	Inches	mm
.002	.05	.575	14.60
.025	.64	.600	15.24
.030	.76	.670	17.02
.060	1.52	.675	17.14
.270	6.86	.720	18.29
.375	9.52	.770	19.56
.400	10.16	1.175	29.84
.470	11.94	1.250	31.75
.475	12.06	1.300	33.02
.500	12.70	1.375	34.92
.570	14.48	1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Capacitor style HVR04.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Conformally coated, molded, or encapsulated.

Lead material: In accordance with MIL-STD-1276.

Capacitance value: See table I.

Capacitance tolerance: J = ±5%, K = ±10%, M = ±20%. See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: 4000 V dc.

Failure rate level: M (1.0%), P (0.1%), R (.01%), or S (.001%).

Dielectric types: BP, BR, and BZ in accordance with MIL-C-49467.

Temperature coefficient: 0 ±30 ppm/°C (BP), ±15% (BR, BZ).

Voltage-temperature limits: In accordance with MIL-C-49467.

Symbol	Capacitance change with reference to +25°C	
	Steps A through D of voltage-temperature limit cycle table of MIL-C-49467. Bias = 0 volt.	Steps E through G of voltage-temperature limit cycle table of MIL-C-49467.
BP	0 ±30 ppm/°C	100% rated voltage: 0 ±30 ppm/°C
BR	±15%	100% rated voltage: +15, -40% 40% rated voltage: +15, -25% 60% rated voltage: +15, -40%
BZ	±15%	

Dissipation factor (25°C): 2.5% maximum (BR and BZ characteristics) or 0.1% maximum (BP characteristic), measured under the same conditions as capacitance.

Resistance to soldering heat: ΔC: -1% to +2% for BP characteristic.
-1% to +6% for BR and BZ characteristics.

Marking: In accordance with MIL-C-49467.

APPLICATION NOTES:

Additional encapsulation is necessary in applications where the possibility of a voltage breakdown between the leads of the capacitor, or the capacitor to another potential, could occur.

Heat sinks on each lead or adequate preheating is required when these capacitors are installed in or removed from circuits by soldering iron.

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TABLE I. Capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467P04180__	18 pF	J, K	BP	D
M49467P04220__	22	J, K	BP	D
M49467P04270__	27	J, K	BP	D
M49467P04330__	33	J, K	BP	D
M49467P04390__	39	J, K	BP	D
M49467P04470__	47	J, K	BP	D
M49467P04560__	56	J, K	BP	D
M49467P04680__	68	J, K	BP	D
M49467P04820__	82	J, K	BP	D
M49467P04101__	100	J, K	BP	D
M49467_04101__	100	K, M	BR, BZ	D
M49467P04121__	120	J, K	BP	D
M49467_04121__	120	K, M	BR, BZ	D
M49467P04151__	150	J, K	BP	D
M49467_04151__	150	K, M	BR, BZ	D
M49467P04181__	180	J, K	BP	D
M49467_04181__	180	K, M	BR, BZ	D
M49467P04221__	220	J, K	BP	D
M49467_04221__	220	K, M	BR, BZ	D
M49467P04271__	270	J, K	BP	D
M49467_04271__	270	K, M	BR, BZ	D
M49467P04331__	330	J, K	BP	D
M49467_04331__	330	K, M	BR, BZ	D
M49467P04391__	390	J, K	BP	D
M49467_04391__	390	K, M	BR, BZ	D
M49467P04471__	470	J, K	BP	D
M49467_04471__	470	K, M	BR, BZ	D
M49467P04561__	560	J, K	BP	D
M49467_04561__	560	K, M	BR, BZ	D
M49467P04681__	680	J, K	BP	D
M49467_04681__	680	K, M	BR, BZ	D
M49467P04821__	820	J, K	BP	D
M49467_04821__	820	K, M	BR, BZ	D
M49467P04102__	1000	J, K	BP	D
M49467_04102__	1000	K, M	BR, BZ	D
M49467P04122__	1200	J, K	BP	E
M49467_04122__	1200	K, M	BR, BZ	D
M49467P04152__	1500	J, K	BP	E
M49467_04152__	1500	K, M	BR, BZ	D
M49467P04182__	1800	J, K	BP	E
M49467_04182__	1800	K, M	BR, BZ	D
M49467P04222__	2200	J, K	BP	F
M49467_04222__	2200	K, M	BR, BZ	D
M49467P04272__	2700	J, K	BP	F
M49467_04272__	2700	K, M	BR, BZ	D
M49467P04332__	3300	J, K	BP	F
M49467_04332__	3300	K, M	BR, BZ	D
M49467P04392__	3900	J, K	BP	G
M49467_04392__	3900	K, M	BR, BZ	D
M49467P04472__	4700	J, K	BP	G
M49467_04472__	4700	K, M	BR, BZ	D

See footnote at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
M49467R04562__	5600 pF	K, M	BR	E
M49467Z04562__	5600	K, M	BZ	D
M49467R04682__	6800	K, M	BR	E
M49467Z04682__	6800	K, M	BZ	D
M49467R04822__	8200	K, M	BR	F
M49467Z04822__	8200	K, M	BZ	E
M49467R04103__	.01 µF	K, M	BR	F
M49467Z04103__	.01	K, M	BZ	E
M49467R04123__	.012	K, M	BR	G
M49467Z04123__	.012	K, M	BZ	F
M49467R04153__	.015	K, M	BR	G
M49467Z04153__	.015	K, M	BZ	F
M49467 04183__	.018	K, M	BR, BZ	G
M49467R04223__	.022	K, M	BR	J
M49467Z04223__	.022	K, M	BZ	G
M49467R04273__	.027	K, M	BR	J
M49467Z04273__	.027	K, M	BZ	G
M49467R04333__	.033	K, M	BR	K
M49467Z04333__	.033	K, M	BZ	J
M49467R04393__	.039	K, M	BR	K
M49467Z04393__	.039	K, M	BZ	J
M49467 04473__	.047	K, M	BR, BZ	K
M49467Z04563__	.056	K, M	BZ	K

1/ The complete PIN shall include additional letter symbols to indicate the characteristic, where applicable, the capacitance tolerance, and the failure rate level.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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